

The empirical recurrence for OEIS A279164, recovered from its tail

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Abstract

OEIS A279164 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 640$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{84} = -3511808 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A279164 is “Number of $n \times 4$ 0..1 arrays with no element equal to a strict majority of its king-move neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

3, 8, 117, 864, 5628, 34764, 203226, 1143396, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Dec 07 2016 08:10:54, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 640$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 640$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1280$ exact terms from $n = 5$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	21	c_{22}	−31889742186	c_{43}	35074545502818	c_{64}	−31164630983562
c_2	−192	c_{23}	65405100150	c_{44}	−94029369842292	c_{65}	23073554424174
c_3	1102	c_{24}	−102821503720	c_{45}	28047006343038	c_{66}	−15412823347526
c_4	−5151	c_{25}	138107882427	c_{46}	−56308299751179	c_{67}	12674042394198
c_5	21915	c_{26}	−304072836051	c_{47}	126148535583495	c_{68}	−8836403135106
c_6	−80658	c_{27}	402753741574	c_{48}	−45374463064150	c_{69}	5653161314420
c_7	262611	c_{28}	−489003417642	c_{49}	80643740380596	c_{70}	−3884061749640
c_8	−819717	c_{29}	1211175433617	c_{50}	−140125738558845	c_{71}	2328935611608
c_9	2356564	c_{30}	−1257448780870	c_{51}	62111828116805	c_{72}	−1265103084520
c_{10}	−6153933	c_{31}	1468854898845	c_{52}	−100464792684900	c_{73}	785555439276
c_{11}	15896799	c_{32}	−4131177915717	c_{53}	133415717185521	c_{74}	−429220770336
c_{12}	−38755379	c_{33}	3113802895495	c_{54}	−69187492951159	c_{75}	154248098496
c_{13}	86192865	c_{34}	−3886722192603	c_{55}	102747012815700	c_{76}	−51442814352
c_{14}	−196345902	c_{35}	11921913822891	c_{56}	−111349922227272	c_{77}	23631300504
c_{15}	424357861	c_{36}	−6185508567318	c_{57}	65391905418118	c_{78}	−9017027688
c_{16}	−826492497	c_{37}	9199626429972	c_{58}	−84324085960743	c_{79}	1584995952
c_{17}	1736557863	c_{38}	−28743434135004	c_{59}	78905675026593	c_{80}	−494177664
c_{18}	−3398705809	c_{39}	10420219415247	c_{60}	−52036752155952	c_{81}	204993728
c_{19}	5847636156	c_{40}	−19234452526020	c_{61}	57102439520595	c_{82}	−42404352
c_{20}	−11832793404	c_{41}	57360728902854	c_{62}	−46441448161626	c_{83}	−3604224
c_{21}	20994425221	c_{42}	−16747937817810	c_{63}	32211479954283	c_{84}	−3511808

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A279164 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = 3511808$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -3511808 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A279164.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.